Supplemental Material:

Bootstrap Inference in the Presence of Bias

by

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This supplemental material contains two appendices. Appendix A describes in detail the conditions and results of the main paper under the special case of asymptotically Gaussian statistics. Appendix B contains details and proofs for the three examples in the main paper, as well as two additional examples. Additional references are included at the end of the supplement.

A Special case: T_n is asymptotically Gaussian

In this section, we specialize Assumptions 1, 2, and 3 to the case where $T_n = \sqrt{n}(\hat{\theta}_n - \theta_0)$ is a normalized parameter estimator whose limiting distribution is normal. We consider the following special case of Assumption 1.

Assumption $1'$ *It holds that $T_n - B_n \to_d N(0, v^2)$, where $v^2 > 0$.*

Assumption $1'$ covers statistics T_n based on asymptotically biased estimators: when $B_n \to_p B$, we have $T_n \to_d N(B, v^2)$, in which case B is the asymptotic bias of $\hat{\theta}_n$. More generally, we can interpret B_n as a bias term that approximates $E(\sqrt{n}(\hat{\theta}_n - \theta_0))$ although B_n does not need to have a limit. Note that Assumption $1'$ obtains from Assumption 1 when we let $\xi_1 \sim N(0, v^2)$ and $G(u) = \Phi(u/v)$.

Let D_n^* denote a bootstrap sample from D_n and let $\hat{\theta}_n^*$ be a bootstrap version of $\hat{\theta}_n$. The bootstrap analogue of T_n is $T_n^* = \sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n)$.

Assumption $2'$ *It holds that (i) $T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v^2)$, and (ii)*

$$\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} \xrightarrow{d} N(0, V), \quad V := (v_{ij}), \quad i, j = 1, 2,$$

where $v_d^2 := v_{11} + v_{22} - 2v_{12} > 0$ with $v_{11} := v^2 > 0$.

Assumption $2'$(i) requires the bootstrap statistic $T_n^* - \hat{B}_n$ to mimic the asymptotic distribution of $T_n - B_n$, as in Assumption 2(i). However, and contrary to Assumption 2(i), here this limiting distribution is the zero mean Gaussian distribution (i.e. $G(u) = \Phi(u/v)$), which means that we can interpret $\hat{B}_n$ as a bootstrap bias correction term; i.e., $\hat{B}_n = E^*(\sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n))$. Assumption $2'$(ii) assumes that $\hat{B}_n - B_n$ is also asymptotically distributed as a zero mean Gaussian random variable (jointly with $T_n - B_n$).[3] An implication of this assumption is that

$$T_n - \hat{B}_n = (T_n - B_n) - (\hat{B}_n - B_n) \xrightarrow{d} N(0, v_d^2), \tag{A.1}$$

[3]In terms of Assumption 2, Assumption $2'$ corresponds to the case where the vector $\xi = (\xi_1, \xi_2)'$ is a multivariate normal distribution with covariance matrix V.

where $v_d^2 := v_{11} + v_{22} - 2v_{12}$. We do not require V to be positive definite; for instance, $v_{22} = 0$ whenever $\hat{B}_n - B_n = o_p(1)$, and in fact V can be rank deficient even when $v_{22} > 0$. However, we do impose the restriction that $v_d^2 > 0$. This ensures that the limiting distribution function of $T_n - \hat{B}_n$, given by $F(u) = \Phi(u/v_d)$, is well-defined and continuous, as assumed in Assumption 2(ii).

Let $\hat{p}_n$ denote the standard bootstrap p-value as defined in Section 3. We then obtain the following.

Corollary A.1 *Under Assumptions 1′ and 2′, $\hat{p}_n \to_d \Phi(m\Phi^{-1}(U_{[0,1]}))$, where $m^2 := v_d^2/v^2$.*

Corollary A.1 follows immediately from Theorem 3.1 when we let $G(u) = \Phi(u/v)$ and $F(u) = \Phi(u/v_d)$. It shows that the asymptotic distribution of $\hat{p}_n$ is uniform only when $m = 1$, or equivalently when $v_d^2 = v^2$. In this case, the difference $\hat{B}_n - B_n$ is $o_p(1)$. When $v_d^2 \neq v^2$, $\hat{B}_n - B_n$ is random even in the limit, implying that the limiting bootstrap distribution function of T_n^* is conditionally random. Although random limit bootstrap measures do not necessarily invalidate bootstrap inference, as discussed by Cavaliere and Georgiev (2020), this is not the case here. However, we can solve the problem of bootstrap invalidity by applying the prepivoting approach or by modifying the test statistic from T_n to $T_n - \hat{B}_n$.

To describe the prepivoting approach, note that the limiting distribution of $\hat{p}_n$ is given by

$$H(u) := \lim P(\hat{p}_n \leq u) = \Phi(m^{-1}\Phi^{-1}(u)).$$

Hence, in this case $\gamma = m$, and a plug-in approach amounts to estimating $m^2 := v_d^2/v^2$, where v^2 and v_d^2 are defined in Assumption 2′. Suppose that $\hat{v}_n^2$ and $\hat{v}_{d,n}^2$ are consistent estimators of v^2 and v_d^2 (i.e., assume that $(\hat{v}_n^2, \hat{v}_{d,n}^2) \to_p (v^2, v_d^2))$ and let $\hat{m}_n^2 := \hat{v}_{d,n}^2/\hat{v}_n^2$. Then, by Corollary 3.2, it immediately follows that

$$\tilde{p}_n = \Phi(\hat{m}_n^{-1}\Phi^{-1}(\hat{p}_n)) \xrightarrow{d} U_{[0,1]}$$

under Assumptions 1′ and 2′. For brevity, we do not formalize this result here.

To describe the double bootstrap modified p-value, $\tilde{p}_n := \hat{H}_n(\hat{p}_n) = P^*(\hat{p}_n^* \leq \hat{p}_n)$, when applied to the special case where T_n satisfies Assumption 1′, we now introduce Assumption 3′.

Assumption 3′ *Let $T_n^{**} = \sqrt{n}(\hat{\theta}_n^{**} - \hat{\theta}_n^*)$ and suppose that (i) $T_n^{**} - \hat{B}_n^* \xrightarrow{d^{**}}_{p^*} N(0, v^2)$, in probability, and (ii) $T_n^* - \hat{B}_n^* \xrightarrow{d^*}_p N(0, v_d^2)$, where v_d^2 is as defined in Assumption 2′(ii).*

Under Assumption 3′(i), the double bootstrap distribution of $T_n^{**} - \hat{B}_n^*$ mimics the distribution of $T_n^* - \hat{B}_n$, where the double bootstrap bias term $\hat{B}_n^* = E^{**}(\sqrt{n}(\hat{\theta}_n^{**} - \hat{\theta}_n^*))$ is asymptotically centered at $\hat{B}_n$ under Assumption 3′(ii). When $v_d^2 \neq v^2$, the double bootstrap bias is not a consistent estimator of $\hat{B}_n$, but that is not needed for the asymptotic validity of the modified double bootstrap p-value $\tilde{p}_n = \hat{H}_n(\hat{p}_n)$ defined in Section 3.

By application of Theorem 3.2, $\tilde{p}_n = \hat{H}_n(\hat{p}_n) \to_d U_{[0,1]}$ under Assumptions 1′, 2′, and 3′. We can also provide a result analogous to Corollary 3.3 under these assumptions. In this case, if closed-form expressions for $\hat{B}_n$ and $\hat{B}_n^*$ are not available, we can approximate these bootstrap expectations by Monte Carlo simulations and then compute $P^*(T_n^* - \hat{B}_n^* \leq T_n - \hat{B}_n)$ as a valid bootstrap p-value. Note, however, that this approach is computationally as intensive as the prepivoting approach based on $\tilde{p}_n$ since it too requires two layers of resampling.

Remark A.1 *In the case of asymptotically Gaussian statistics discussed in this section, the more general Assumptions 4 and 5 simplify straightforwardly. In Assumption 2′(i) we assume that* $T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v_s^2)$ *and in Assumption 3′(i) that* $T_n^{**} - \hat{B}_n^* \xrightarrow{d^{**}}_{p^*} N(0, v_s^2)$*, in probability, for some* $v_s^2 > 0$*, while the rest of Assumptions 1′–3′ are unchanged. The results of this section continue to apply under these more general conditions, replacing* $G(u) = \Phi(u/v)$ *with* $J(u) = \Phi(u/v_s)$ *and consequently defining* $m := v_d^2/v_s^2$*.*

Remark A.2 *Contrary to Beran (1987, 1988), in our context the first level of prepivoting, e.g., by the double bootstrap, is used to obtain an asymptotically valid bootstrap p-value. Therefore, inference based on* $\tilde{p}_n$ *does not necessarily provide an asymptotic refinement over inference based on an asymptotic approach that does not require the bootstrap. Nevertheless, the Monte Carlo results in Table B.1 below seem to suggest an asymptotic refinement for the double bootstrap, at least for the non-parametric bootstrap scheme. In the special case where the bias term* B_n *is of sufficiently small order, the arguments in Beran (1987, 1988) apply, and an asymptotic refinement can be obtained. We also conjecture that, in the general case, an asymptotic refinement could be obtained by further iterating the bootstrap.*

B Examples with details

B.1 Inference after model averaging

In this section we first provide the regularity conditions required in Lemmas 4.1 and 4.2, and then we give the proofs of the lemmas. We subsequently provide some brief Monte Carlo evidence. Finally, at the end of the section, we provide regularity conditions for the extension to the pairs bootstrap and a proof of the associated Lemma 4.3.

B.1.1 Assumptions and notation

We impose the following conditions.

Assumption MA *(i)* $\varepsilon_t|W \sim i.i.d.(0,\sigma^2)$, *where* $W := (x, Z)$*; (ii)* $S_{WW} \rightarrow_p \Sigma_{WW}$ *with* $\text{rank}(\Sigma_{WW}) = q+1$*; (iii)* $n^{1/2}S_{W\varepsilon} \rightarrow_d N(0,\Omega)$ *with* $\Omega := \sigma^2\Sigma_{WW}$.

Remark B.1 *We assume that the weights* ω *are fixed and independent of* n*. A popular example in forecasting is to use equal weighting. We could allow for stochastic weights as long as these are constant in the limit. This would be the case, for example, when the weights are based on moments that can be consistently estimated.*

To proceed, we introduce the following notation. First, partition Σ_{WW} according to W,

$$\Sigma_{WW} := \begin{pmatrix} \Sigma_{xx} & \Sigma_{xZ} \\ \Sigma_{Zx} & \Sigma_{ZZ} \end{pmatrix}.$$

Let $\Sigma_{xZ_m} := \Sigma_{xZ}R_m$, $\Sigma_{Z_mZ_m} := R_m'\Sigma_{ZZ}R_m$, $\Sigma_{xx.Z_m} := \Sigma_{xx} - \Sigma_{xZ}R_m(R_m'\Sigma_{ZZ}R_m)^{-1}R_m'\Sigma_{Zx}$, and $\Sigma_{xZ.Z_m} := \Sigma_{xZ} - \Sigma_{xZ}R_m(R_m'\Sigma_{ZZ}R_m)^{-1}R_m'\Sigma_{ZZ}$. Also let $A_n := \sum_{m=1}^{M}\omega_m S_{xx.Z_m}^{-1}n^{-1}x'M_{Z_m}$, where $M_{Z_m} := I_n - Z_m(Z_m'Z_m)^{-1}Z_m'$, such that $A_nZ = Q_n$. With this notation,

$$\tilde{\beta}_n = A_ny = A_nx\beta + Q_n\delta + A_n\varepsilon = \beta + Q_n\delta + A_n\varepsilon, \tag{B.1}$$

$$\tilde{\beta}_n^* = A_ny^* = \hat{\beta}_n + Q_n\hat{\delta}_n + A_n\varepsilon^*. \tag{B.2}$$

Finally, define

$$\bar{d}'_{M,n} := \sum_{m=1}^{M} \omega_m S_{xx.Z_m}^{-1}(1, -S_{xZ_m} S_{Z_m Z_m}^{-1} R'_m),$$

$$\bar{b}'_{M,n} := \sum\nolimits_{m=1}^{M} \omega_m S_{xx.Z_m}^{-1} S_{xZ.Z_m} S_{ZZ.x}^{-1}(-S_{Zx} S_{xx}^{-1}, I_q),$$

and let $\bar{d}'_M$ and $\bar{b}'_M$ denote their probability limits, which exist and are well-defined under Assumption MA.

B.1.2 Proofs of lemmas

Proof of Lemma 4.1. We first verify Assumption 1 (or equivalently, Assumption 1′). Using (B.1) we can write $T_n = B_n + \xi_{1,n}$ with

$$\xi_{1,n} := n^{1/2} A_n \varepsilon = n^{1/2} \sum_{m=1}^{M} \omega_m S_{xx.Z_m}^{-1} n^{-1} x' M_{Z_m} \varepsilon = n^{1/2} \sum_{m=1}^{M} \omega_m S_{xx.Z_m}^{-1} S_{x\varepsilon.Z_m}.$$

Then

$$\begin{aligned} S_{x\varepsilon.Z_m} &= n^{-1} x' M_{Z_m} \varepsilon = n^{-1}(x'\varepsilon - x' Z_m (Z'_m Z_m)^{-1} R'_m Z'\varepsilon) \\ &= (1, -S_{xZ_m}(S_{Z_m Z_m})^{-1} R'_m) S_{W\varepsilon} =: \hat{d}'_m S_{W\varepsilon}, \end{aligned}$$

so that

$$\xi_{1,n} = \sum_{m=1}^{M} \omega_m S_{xx.Z_m}^{-1} \hat{d}'_m n^{1/2} S_{W\varepsilon} = \bar{d}'_{M,n} n^{1/2} S_{W\varepsilon}.$$

Hence, $\xi_{1,n} \to_d N(0, v^2)$ with $v^2 := \bar{d}'_M \Omega \bar{d}_M$.

Next, we verify Assumption 2 (or Assumption 2′). From (B.2) we write $T_n^* = \hat{B}_n + \xi_{1,n}^*$ with $\xi_{1,n}^* := n^{1/2} A_n \varepsilon^* \sim N(0, \hat{\sigma}_n^2 A_n A'_n)$, conditional on D_n. Part (i) now follows straightforwardly because $\hat{\sigma}_n^2 \to_p \sigma^2$ and $A_n A'_n = \bar{d}'_{M,n} S_{WW} \bar{d}_{M,n} \to_p \bar{d}'_M \Sigma_{WW} \bar{d}_M$. To prove Part (ii), note that

$$n^{1/2}(\hat{\delta}_n - \delta) = S_{ZZ.x}^{-1} S_{Z\varepsilon.x} = S_{ZZ.x}^{-1}(-S_{Zx} S_{xx}^{-1}, I_q) n^{1/2} S_{W\varepsilon},$$

from which it follows that

$$\hat{B}_n - B_n = Q_n n^{1/2}(\hat{\delta}_n - \delta) = Q_n S_{ZZ.x}^{-1}(-S_{Zx} S_{xx}^{-1}, I_q) n^{1/2} S_{W\varepsilon} = \bar{b}'_{M,n} n^{1/2} S_{W\varepsilon}.$$

Hence,

$$\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} = \begin{pmatrix} \bar{d}'_{M,n} \\ \bar{b}'_{M,n} \end{pmatrix} n^{-1/2} W'\varepsilon \xrightarrow{d} N(0,V), \quad V = \begin{pmatrix} \bar{d}'_M \Omega \bar{d}_M & \bar{d}'_M \Omega \bar{b}_M \\ \bar{b}'_M \Omega \bar{d}_M & \bar{b}'_M \Omega \bar{b}_M \end{pmatrix},$$

which completes the proof. □

Proof of Lemma 4.2. First note that $\tilde{\beta}_n^{**} = A_n y^{**} = A_n x \hat{\beta}_n^* + A_n Z \hat{\delta}_n^* + A_n \varepsilon^{**}$. It follows that

$$T_n^{**} := n^{1/2}(\tilde{\beta}_n^{**} - \hat{\beta}_n^*) = \hat{B}_n^* + n^{1/2} A_n \varepsilon^{**},$$

where $\hat{B}_n^* := n^{1/2} Q_n \hat{\delta}_n^*$ and $\xi_{1,n}^{**} := n^{1/2} A_n \varepsilon^{**} \sim N(0, \hat{\sigma}_n^{*2} A_n A_n')$, conditional on (D_n, D_n^*). The conditions in Assumption 3(i) or 3′(i) now follows as in Part (i) of the previous proof because $\hat{\sigma}_n^{*2} \xrightarrow{p^*}_p \sigma^2$. For Assumption 3(ii) or 3′(ii) we consider the joint convergence of $(T_n^* - \hat{B}_n, \hat{B}_n^* - \hat{B}_n)'$. By noticing that

$$n^{1/2}(\hat{\delta}_n^* - \hat{\delta}_n) = S_{ZZ.x}^{-1} S_{Z\varepsilon^*.x} = S_{ZZ.x}^{-1}(-S_{Zx}S_{xx}^{-1}, I_q) n^{1/2} S_{W\varepsilon^*},$$

it follows that

$$\begin{pmatrix} T_n^* - \hat{B}_n \\ \hat{B}_n^* - \hat{B}_n \end{pmatrix} = \begin{pmatrix} \bar{d}'_{M,n} \\ \bar{b}'_{M,n} \end{pmatrix} n^{1/2} S_{W\varepsilon^*} \sim N(0, \hat{V}_n),$$

conditional on D_n, where

$$\hat{V}_n = \hat{\sigma}_n^2 \begin{pmatrix} \bar{d}'_{M,n} S_{WW} \bar{d}_{M,n} & \bar{d}'_{M,n} S_{WW} \bar{b}_{M,n} \\ \bar{b}'_{M,n} S_{WW} \bar{d}_{M,n} & \bar{b}'_{M,n} S_{WW} \bar{b}_{M,n} \end{pmatrix} \xrightarrow{p} V.$$

The desired result follows. □

B.1.3 A small Monte Carlo experiment

In Table B.1 we present the results of a small Monte Carlo simulation experiment to illustrate the above results numerically. We generate the data from the regression model (2.1) with sample sizes $n = 10, 20, 40$. The regressors x_t and z_t are both scalar and multivariate normally distributed with unit variances and correlation 0.7, and the errors are either standard normal, t_3, or χ_1^2 distributed. The true values are $\beta = \bar{\beta} + an^{-1/2}$ with $\bar{\beta} = 1$ and $\delta = 1$ (the results are invariant to $\bar{\beta}$ and δ because we use the unrestricted estimates to construct the bootstrap samples). We test the null hypothesis $\mathsf{H}_0 : \beta = \bar{\beta}$

against a left-sided alternative. Results for right-tailed and two-tailed tests are analogous to those presented here for left-tailed tests. The case $a = 0$ corresponds to rejection frequencies under the null, and $a = -1, -2, -4$ corresponds to rejection frequencies under local alternatives. The estimator puts weight $\omega_1 = 1/2$ on the short model that includes only x (and a constant term) and weight $\omega_2 = 1/2$ on the long model that includes both regressors (and a constant term). We consider two bootstrap schemes. The first is the parametric bootstrap scheme, where $\varepsilon_t^* \sim$ i.i.d.$N(0, 1)$, which is denoted as "par." in the table. The second is the non-parametric bootstrap scheme, where ε_t^* is resampled independently from the (centered) residuals from the long regression, which is denoted as "non-par." Results are based on 10,000 Monte Carlo simulations and $B = 999$ bootstrap replications.

First consider the case $a = 0$. The simulation outcomes in Table B.1 clearly illustrate our theoretical results. The standard bootstrap p-value, $\hat{p}_n$, is much larger than the nominal level of the test. The plug-in modified p-value, $\tilde{p}_{n,p}$, is close to the nominal level for the parametric bootstrap scheme, but is still over-sized for the non-parametric scheme with the smaller sample sizes. Finally, the double bootstrap modified p-value, $\tilde{p}_{n,d}$, is nearly perfectly sized throughout the table.

Table B.1 for $a = -1, -2, -4$ clearly shows nontrivial power, which increases as a increases. The discrepancies in finite-sample power are due to differences in size. For example, consider the standard parametric bootstrap with 5% nominal level and normal errors (top left of the table). It has finite-sample size very close to 10%. Comparing this with our modified bootstrap test with nominal size 10% (towards the right in the same panel of the table), we see that the finite-sample powers are nearly identical.

B.1.4 Extension to the pairs bootstrap

In addition to Assumption MA we also impose the following conditions.

Assumption MA$_2$ *With $w_t := (x_t, z_t)'$ it holds that (i) $\sup_t E\,\|w_t\|^4 < \infty$, $E\varepsilon_t^4 < \infty$; (ii) $n^{-1}\sum_{t=1}^n x_t^2\varepsilon_t^2 \to_p \sigma^2\Sigma_{xx}$, $n^{-1}\sum_{t=1}^n x_t^2 w_t w_t' \to_p \Sigma_r > 0$, and $n^{-1}\sum_{t=1}^n x_t^2 w_t \varepsilon_t \to_p 0$.*

Table B.1: Simulated rejection frequencies (%) of bootstrap tests

			5% nominal level						10% nominal level					
			par.			non-par.			par.			non-par.		
dist.	a	n	$\hat{p}_n$	$\tilde{p}_{n,p}$	$\tilde{p}_{n,d}$	$\hat{p}_n$	$\tilde{p}_{n,p}$	$\tilde{p}_{n,d}$	$\hat{p}_n$	$\tilde{p}_{n,p}$	$\tilde{p}_{n,d}$	$\hat{p}_n$	$\tilde{p}_{n,p}$	$\tilde{p}_{n,d}$
N	0	10	10.1	5.0	5.0	16.2	11.2	6.3	15.9	10.0	10.0	21.5	16.2	10.5
		20	9.7	5.0	5.1	12.6	7.8	5.4	15.1	9.8	9.8	18.2	12.9	10.4
		40	9.8	5.1	5.2	10.5	5.8	4.9	15.4	10.1	10.2	16.5	11.0	9.8
	-1	10	25.5	15.9	16.0	34.0	26.1	16.2	35.7	25.9	25.8	42.3	34.3	24.8
		20	26.0	16.5	16.7	30.1	21.0	15.9	36.5	26.9	26.6	40.0	30.9	26.1
		40	27.4	17.7	17.9	29.3	19.4	16.9	37.8	28.4	28.4	38.9	30.0	27.6
	-2	10	47.7	35.6	35.7	57.0	47.5	33.2	58.4	48.5	48.1	64.9	57.4	45.7
		20	51.6	38.3	38.3	56.3	44.0	36.2	62.5	52.1	52.3	65.9	56.9	51.4
		40	52.5	39.9	39.8	54.8	43.0	39.1	63.9	53.6	53.8	64.9	55.5	52.8
	-4	10	84.9	75.6	75.6	88.2	82.5	71.6	90.1	84.5	84.3	91.9	87.9	81.2
		20	90.5	82.9	82.7	91.5	85.5	80.2	94.2	90.3	90.0	94.4	91.3	88.8
		40	91.7	85.4	85.3	92.5	87.0	84.7	95.3	92.2	92.0	95.8	92.7	91.7
t_3	0	10	7.3	3.7	3.8	15.6	10.8	5.8	12.0	7.3	7.2	21.5	15.8	10.2
		20	7.5	4.1	4.2	13.2	8.1	5.6	12.7	7.6	7.9	19.0	13.4	10.9
		40	7.5	3.8	3.9	10.5	5.7	4.9	12.8	7.8	7.8	16.6	10.8	9.6
	-1	10	20.9	12.0	11.9	39.4	30.6	19.8	31.7	21.4	21.3	47.7	39.8	29.5
		20	23.3	13.1	13.3	35.2	25.3	19.3	34.2	23.9	24.0	45.0	36.3	31.1
		40	24.6	14.7	14.7	31.8	21.4	19.2	35.6	25.3	25.3	42.3	32.9	30.5
	-2	10	47.5	32.2	32.2	65.2	56.6	42.8	60.3	47.7	47.6	72.7	65.4	55.1
		20	51.4	36.7	37.0	63.7	52.3	45.1	64.4	52.4	52.4	72.5	63.9	59.1
		40	52.8	38.1	38.3	60.8	47.8	44.6	65.6	53.9	53.7	70.9	61.7	58.9
	-4	10	87.7	78.1	77.9	91.3	86.9	78.6	92.1	87.3	87.2	94.1	91.2	85.9
		20	91.8	85.0	84.9	92.6	88.1	84.2	95.1	91.6	91.6	95.3	92.8	91.0
		40	93.2	87.7	87.6	93.2	88.2	86.8	96.1	93.3	93.3	96.0	93.3	92.5
χ^2_1	0	10	8.3	4.7	4.7	16.0	10.7	5.8	12.6	8.0	8.0	21.5	16.2	9.9
		20	8.5	4.9	4.9	12.2	7.0	5.0	13.5	8.6	8.6	18.1	12.4	9.8
		40	9.2	4.9	4.8	10.9	6.1	5.3	14.8	9.7	9.5	17.1	11.2	10.1
	-1	10	21.1	12.6	12.6	41.9	33.2	22.5	30.9	21.7	21.2	50.1	42.0	31.9
		20	23.4	14.3	14.3	35.1	25.1	19.7	33.6	24.0	24.1	45.2	35.9	31.2
		40	25.5	15.8	15.9	31.7	21.2	19.1	36.2	26.6	26.7	42.2	32.7	30.4
	-2	10	46.9	31.3	31.5	65.2	57.2	45.3	60.6	47.6	47.6	72.0	65.4	55.8
		20	51.2	36.3	36.4	62.2	51.4	44.3	64.3	52.4	52.5	71.3	62.9	57.9
		40	53.9	39.2	39.1	59.4	46.9	43.9	65.2	55.1	54.9	69.9	60.4	57.8
	-4	10	87.2	78.5	78.3	88.8	84.3	76.6	91.5	86.6	86.4	91.8	88.6	83.2
		20	91.1	84.7	84.7	90.6	84.6	80.4	94.2	91.0	90.8	93.9	90.5	88.4
		40	92.6	86.8	86.8	91.8	86.7	85.2	95.6	92.7	92.5	94.8	92.0	91.0

Notes: $\hat{p}_n$ denotes the standard bootstrap; $\tilde{p}_{n,p}$ and $\tilde{p}_{n,d}$ denote the modified bootstrap using the plug-in and the double bootstrap methods, respectively. The parametric bootstrap scheme, where $\varepsilon_t^* \sim$ i.i.d.$N(0,1)$, is denoted as "par." and the non-parametric bootstrap scheme, where ε_t^* is re-sampled independently from the long regression (centered) residuals, is denoted as "non-par." The ε_t's are i.i.d. draws from (standardized) N, t_3, and χ^2_1 distributions. The parameter a denotes the drift under the local alternative $\beta_0 = \bar{\beta} + an^{-1/2}$. Results are based on 10,000 Monte Carlo simulations and $B = 999$ bootstrap replications for each level.

Proof of Lemma 4.3. We first prove that

$$S_{W^*W^*} - S_{WW} \xrightarrow{p^*}_p 0, \tag{B.3}$$

$$S_n^* := \begin{pmatrix} n^{1/2} S_{x^*\varepsilon^*} \\ n^{1/2}(S_{x^*z^*} - S_{xz}) \\ n^{1/2}(S_{x^*x^*} - S_{xx}) \end{pmatrix} \xrightarrow{d^*}_p N(0, \Sigma_s), \quad \Sigma_s = \begin{pmatrix} \sigma^2 \Sigma_{xx} & 0 \\ 0 & \Sigma_r \end{pmatrix}. \tag{B.4}$$

Here, (B.3) follows by straightforward application of Chebyshev's LLN.

To prove (B.4), we first compute the mean and variance of S_n^*. Note that the mean of S_n^* is zero by construction; for example, $E^*(n^{1/2} S_{x^*\varepsilon^*}) = n^{-1/2} \sum_{t=1}^n E^*(x_t^* \varepsilon_t^*) = n^{1/2} S_{x\hat{\varepsilon}} = 0$ by the OLS first-order condition. In addition,

$$\mathrm{Var}^*(n^{1/2} S_{x^*\varepsilon^*}) = n^{-1} \sum_{t=1}^n E^*(x_t^{*2} \varepsilon_t^{*2}) = n^{-1} \sum_{t=1}^n x_t^2 \hat{\varepsilon}_t^2 \xrightarrow{p} \sigma^2 \Sigma_{xx}$$

under Assumptions MA and MA_2. Similarly, letting

$$\begin{pmatrix} n^{1/2}(S_{x^*z^*} - S_{xz}) \\ n^{1/2}(S_{x^*x^*} - S_{xx}) \end{pmatrix} = n^{1/2}(S_{x^*W^*} - S_{xW}),$$

we find that

$$\begin{aligned} \mathrm{Var}^*(n^{1/2}\,(S_{x^*W^*} - S_{xW})) &= n^{-1} \sum_{t=1}^n (x_t w_t - E^*(x_t^* w_t^*))(x_t w_t - E^*(x_t^* w_t^*))' \\ &= n^{-1} \sum_{t=1}^n x_t^2 w_t w_t' - S_{xW} S_{Wx} \xrightarrow{p} \Sigma_r - \Sigma_{xW} \Sigma_{Wx}. \end{aligned}$$

Note also that the covariance between $n^{1/2} S_{x^*\varepsilon^*}$ and $n^{1/2}(S_{x^*W^*} - S_{xW})$ is zero because

$$\begin{aligned} E^*(n S_{x^*\varepsilon^*} S_{x^*W^*}) &= n^{-1} E^* \left(\sum_{t=1}^n x_t^* \varepsilon_t^* \sum_{s=1}^n x_s^* w_s^* \right) = n^{-1} E^* \left(\sum_{t=1}^n x_t^{*2} w_t^* \varepsilon_t^* \right) \\ &= E^*(x_t^{*2} w_t^* \varepsilon_t^*) = n^{-1} \sum_{t=1}^n x_t^2 w_t \hat{\varepsilon}_t \xrightarrow{p} 0 \end{aligned}$$

by Assumption MA_2(ii). Thus, we have shown that $E^*(S_n^*) = 0$ and $E^*(S_n^* S_n^{*\prime}) \to_p \Sigma_s$. The result (B.4) now follows because the stated moment conditions imply the Lindeberg condition by standard arguments.

Next we can write

$$T_n^* - \hat{B}_n = n^{1/2} S_{x^*x^*}^{-1} S_{x^*\varepsilon^*} + B_n^* - \hat{B}_n,$$

where

$$B_n^* - \hat{B}_n = (S_{x^*x^*}^{-1} S_{x^*z^*} - S_{xx}^{-1} S_{xz}) n^{1/2} \hat{\delta}_n.$$

Adding and subtracting appropriately, we can write this difference as

$$B_n^* - \hat{B}_n = n^{1/2}(S_{x^*x^*}^{-1} S_{x^*z^*} - S_{xx}^{-1} S_{xz})\delta + (S_{x^*x^*}^{-1} S_{x^*z^*} - S_{xx}^{-1} S_{xz}) n^{1/2}(\hat{\delta}_n - \delta),$$

where $n^{1/2}(\hat{\delta}_n - \delta)$ is $O_p(1)$ by a central limit theorem and $S_{x^*x^*}^{-1} S_{x^*z^*} - S_{xx}^{-1} S_{xz} = o_{p^*}(1)$, in probability, by (B.3). The first term in $B_n^* - \hat{B}_n$ can be written as

$$\begin{aligned}
&S_{x^*x^*}^{-1} n^{1/2}(S_{x^*z^*} - S_{xz})\delta - S_{x^*x^*}^{-1} S_{xx}^{-1}(S_{x^*x^*} - S_{xx}) n^{1/2} S_{xz}\delta \\
&= \delta(\Sigma_{xx}^{-1}, -\Sigma_{xx}^{-2}\Sigma_{xz}) \begin{pmatrix} n^{1/2}(S_{x^*z^*} - S_{xz}) \\ n^{1/2}(S_{x^*x^*} - S_{xx}) \end{pmatrix} + o_{p^*}(1),
\end{aligned}$$

in probability, by application of (B.3) and Assumption MA(ii). It follows that

$$\begin{aligned}
T_n^* - \hat{B}_n &= S_{x^*x^*}^{-1} n^{1/2} S_{x\varepsilon}^* + \delta(\Sigma_{xx}^{-1}, -\Sigma_{xx}^{-2}\Sigma_{xz}) \begin{pmatrix} n^{1/2}(S_{x^*z^*} - S_{xz}) \\ n^{1/2}(S_{x^*x^*} - S_{xx}) \end{pmatrix} + o_{p^*}(1) \\
&= (\Sigma_{xx}^{-1}, \Sigma_{xx}^{-1}\delta, -\Sigma_{xx}^{-2}\Sigma_{xz}\delta) S_n^* + o_{p^*}(1),
\end{aligned}$$

in probability. The required result now follows from (B.4) because

$$\begin{aligned}
&(\Sigma_{xx}^{-1}, \Sigma_{xx}^{-1}\delta, -\Sigma_{xx}^{-2}\Sigma_{xz}\delta) \begin{pmatrix} \Sigma_s & 0 \\ 0 & \Sigma_r \end{pmatrix} (\Sigma_{xx}^{-1}, \Sigma_{xx}^{-1}\delta, -\Sigma_{xx}^{-2}\Sigma_{xz}\delta)' \\
&= \Sigma_{xx}^{-1}\Sigma_s\Sigma_{xx}^{-1} + d_r(\delta)'\Sigma_r d_r(\delta) = v^2 + \kappa^2,
\end{aligned}$$

which completes the proof upon noting that $\Sigma_s = \sigma^2\Sigma_{xx}$ implies $v^2 = \sigma^2\Sigma_{xx}^{-1}$. □

B.2 Ridge regression

B.2.1 Assumptions and notation

As in Fu and Knight (2000) we assume the following.

Assumption RE *(i)* $\varepsilon_t \sim i.i.d.(0,\sigma^2)$; *(ii)* $\max_{t=1,\ldots,n} x_t'x_t = o(n)$; *(iii)* S_{xx} *is nonsingular for any* n *and converges to a positive definite matrix,* Σ_{xx}; *(iv)* $\theta = \delta n^{-1/2}$; *and (v)* $n^{-1}c_n \to c_0 \geq 0$.

For the bootstrap we will also need the following.

Assumption RE$_2$ *Assumption RE holds with (ii) replaced by (ii') $\max_{t=1,\ldots,n} x_t'x_t = o(n^{1/2})$ and with the additional condition (vi) $E\varepsilon_t^4 < \infty$.*

Finally, we define

$$V = \sigma^2 \begin{pmatrix} g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}\tilde{\Sigma}_{xx}^{-1}g & -c_0 g'\tilde{\Sigma}_{xx}^{-1}\tilde{\Sigma}_{xx}^{-1}g \\ -c_0 g'\tilde{\Sigma}_{xx}^{-1}\tilde{\Sigma}_{xx}^{-1}g & c_0^2 g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}^{-1}\tilde{\Sigma}_{xx}^{-1}g \end{pmatrix}$$

where $v^2 := v_{11}$, and it holds that

$$m^2 := \frac{v_{11}+v_{22}-2v_{12}}{v_{11}} = \frac{g'\Sigma_{xx}^{-1}g}{g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}\tilde{\Sigma}_{xx}^{-1}g}, \tag{B.5}$$

where the last equality is derived in the proof of Lemma 4.4.

B.2.2 Proofs of lemmas

Proof of Lemma 4.4 and derivation of (B.5). The result follows by showing that

$$\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} = \begin{pmatrix} g'\tilde{S}_{xx}^{-1} \\ -c_n n^{-1} g'\tilde{S}_{xx}^{-1}S_{xx}^{-1} \end{pmatrix} n^{1/2}S_{x\varepsilon} \xrightarrow{d} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \sim N(0,V), \quad V = (v_{ij}), \tag{B.6}$$

and that

$$T_n^* - B_n^* = T_n^* - \hat{B}_n + o_{p^*}(1) \xrightarrow{d^*}_p N(0,v^2). \tag{B.7}$$

To prove (B.6) we first notice that, since $c_n n^{-1} \to c_0$, under Assumption RE we have that $n^{1/2}S_{x\varepsilon} \to_d$

$N(0, \sigma^2 \Sigma_{xx})$ and hence

$$\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} = (I_2 \otimes g' \tilde{S}_{xx}^{-1}) \begin{pmatrix} I_p \\ -n^{-1} c_n S_{xx}^{-1} \end{pmatrix} n^{1/2} S_{x\varepsilon}$$

$$\xrightarrow{d} (I_2 \otimes g' \tilde{\Sigma}_{xx}^{-1}) N \left(0, \sigma^2 \begin{pmatrix} \Sigma_{xx} & -c_0 I_p \\ -c_0 I_p & c_0^2 \Sigma_{xx}^{-1} \end{pmatrix} \right) \sim N(0, V), \tag{B.8}$$

$$V = \sigma^2 \begin{pmatrix} g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx} \tilde{\Sigma}_{xx}^{-1} g & -c_0 g' \tilde{\Sigma}_{xx}^{-1} \tilde{\Sigma}_{xx}^{-1} g \\ -c_0 g' \tilde{\Sigma}_{xx}^{-1} \tilde{\Sigma}_{xx}^{-1} g & c_0^2 g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx}^{-1} \tilde{\Sigma}_{xx}^{-1} g \end{pmatrix}.$$

This immediately implies that m^2 in (B.5) is given by

$$m^2 = \frac{g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx} \tilde{\Sigma}_{xx}^{-1} g + 2c_0 g' \tilde{\Sigma}_{xx}^{-1} \tilde{\Sigma}_{xx}^{-1} g + c_0^2 g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx}^{-1} \tilde{\Sigma}_{xx}^{-1} g}{g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx} \tilde{\Sigma}_{xx}^{-1} g}. \tag{B.9}$$

The numerator of m^2 in (B.9) can be written as

$$g' \tilde{\Sigma}_{xx}^{-1} (\Sigma_{xx} + 2c_0 I_p + c_0^2 \Sigma_{xx}^{-1}) \tilde{\Sigma}_{xx}^{-1} g = g' \tilde{\Sigma}_{xx}^{-1} (\tilde{\Sigma}_{xx} \Sigma_{xx}^{-1} \tilde{\Sigma}_{xx}) \tilde{\Sigma}_{xx}^{-1} g = g' \Sigma_{xx}^{-1} g,$$

and hence (B.5) follows.

To prove (B.7) we note that $T_n^* - \hat{B}_n = \xi_{1,n}^* + B_n^* - \hat{B}_n$, where $\xi_{1,n}^* := n^{1/2} g' \tilde{S}_{x^*x^*}^{-1} S_{x^*\varepsilon^*}$ and

$$\begin{aligned} B_n^* - \hat{B}_n &= -c_n n^{-1/2} g' \tilde{S}_{x^*x^*}^{-1} \hat{\theta}_n + c_n n^{-1/2} g' \tilde{S}_{xx}^{-1} \hat{\theta}_n \\ &= -c_n n^{-1} g' (\tilde{S}_{x^*x^*}^{-1} - \tilde{S}_{xx}^{-1}) n^{1/2} (\hat{\theta}_n - \theta) - c_n n^{-1} g' (\tilde{S}_{x^*x^*}^{-1} - \tilde{S}_{xx}^{-1}) \delta, \end{aligned}$$

such that $B_n^* - \hat{B}_n \xrightarrow{p^*}_p 0$ if $\tilde{S}_{x^*x^*}^{-1} - \tilde{S}_{xx}^{-1} \xrightarrow{p^*}_p 0$. Because $||\tilde{S}_{xx}^{-1}|| = O(1)$ under the stated assumptions, it follows that $||\tilde{S}_{x^*x^*}^{-1} - \tilde{S}_{xx}^{-1}||$ has the same rate as $||\tilde{S}_{x^*x^*} - \tilde{S}_{xx}||$. Thus, $\tilde{S}_{x^*x^*} - \tilde{S}_{xx} = S_{x^*x^*} - S_{xx} = n^{-1} \sum_{t=1}^n x_t^* x_t^{*\prime} - E^*(x_t^* x_t^{*\prime}) \xrightarrow{p^*}_p 0$ by a straightforward application of Chebyshev's LLN using that $\max_t x_t' x_t = o(n^{1/2})$ by Assumption RE_2(ii').

The proof is completed by showing that $\xi_{1,n}^*$ satisfies the bootstrap central limit theorem. By the above results it holds that $\xi_{1,n}^* = n^{1/2} g' \tilde{\Sigma}_{xx}^{-1} S_{x^*\varepsilon^*} + o_{p^*}(1)$, so it is only required to analyze the term $n^{1/2} g' \tilde{\Sigma}_{xx}^{-1} S_{x^*\varepsilon^*} = n^{1/2} S_{\tilde{x}^*\varepsilon^*}$, where $\tilde{x}_t^* := g' \tilde{\Sigma}_{xx}^{-1} x_t^*$. First, we have $E^*(n^{1/2} S_{\tilde{x}^*\varepsilon^*}) = g' \tilde{\Sigma}_{xx}^{-1} E^*(n^{1/2} S_{x^*\varepsilon^*}) =$

$n^{1/2}g'\tilde{\Sigma}_{xx}^{-1}S_{x\hat{\varepsilon}} = 0$. Second, with $\tilde{x}_t := g'\tilde{\Sigma}_{xx}^{-1}x_t$,

$$\begin{aligned}\mathrm{Var}^*(n^{1/2}S_{\tilde{x}^*\varepsilon^*}) &= n^{-1}\sum_{t=1}^{n}\tilde{x}_t^2\hat{\varepsilon}_t^2 = n^{-1}\sum_{t=1}^{n}\tilde{x}_t^2(\hat{\varepsilon}_t^2-\sigma^2+\sigma^2) \\ &= \sigma^2 g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}\tilde{\Sigma}_{xx}^{-1}g + n^{-1}\sum_{t=1}^{n}\tilde{x}_t^2(\varepsilon_t^2-\sigma^2)+o_p(1).\end{aligned}$$

Because ε_t is i.i.d. and $\tilde{x}_t^2$ is non-stochastic, a sufficient condition for $n^{-1}\sum_{t=1}^{n}\tilde{x}_t^2(\varepsilon_t^2-\sigma^2)\to_p 0$ is that $\lambda_{\min}(\sum_{t=1}^{n}\tilde{x}_t^2)\to\infty$, where $\lambda_{\min}(\cdot)$ denotes the minimum eigenvalue of the argument, and this is implied by $n^{-1}\sum_{t=1}^{n}\tilde{x}_t^2\to g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}\tilde{\Sigma}_{xx}^{-1}g>0$.

Third, we check Lindeberg's condition, where we set $s_n^2 := nS_{\tilde{x}\tilde{x}}$. For $\epsilon>0$ it holds that

$$\begin{aligned}\frac{1}{s_n^2}\sum_{t=1}^{n}E^*(\tilde{x}_t^{*2}\varepsilon_t^{*2}\mathbb{I}_{\{|\tilde{x}_t^*\varepsilon_t^*|>\epsilon s_n\}}) &= \frac{1}{S_{\tilde{x}\tilde{x}}}E^*(\tilde{x}_t^{*2}\varepsilon_t^{*2}\mathbb{I}_{\{(\tilde{x}_t^*\varepsilon_t^*)^2>\epsilon^2 nS_{\tilde{x}\tilde{x}}\}}) \\ &\le \frac{1}{\epsilon^2 nS_{\tilde{x}\tilde{x}}^2}E^*(\tilde{x}_t^{*4}\varepsilon_t^{*4}) \\ &= \frac{1}{\epsilon^2 n^2 S_{\tilde{x}\tilde{x}}^2}\sum_{t=1}^{n}\tilde{x}_t^4\hat{\varepsilon}_t^4 \le \frac{n^{-1}\max_t\tilde{x}_t^4}{\epsilon^2 S_{\tilde{x}\tilde{x}}^2}\frac{1}{n}\sum_{t=1}^{n}\hat{\varepsilon}_t^4 \xrightarrow{p} 0\end{aligned}$$

because $n^{-1}\max_t\tilde{x}_t^4 = o(1)$ and ε_t has bounded fourth-order moment. □

Proof of Lemma 4.5. The proof follows closely the proofs of Lemma 4.4 and is omitted for brevity.□

B.3 Nonparametric regression

B.3.1 Assumptions and notation

We impose the following conditions.

Assumption NP *(i)* $\varepsilon_t\sim$ *i.i.d.*$(0,\sigma^2)$*; (ii)* $E|\varepsilon_t|^{2+\delta}<\infty$*; (iii)* $\beta:[0,1]\to\mathbb{R}$ *is three times continuously differentiable with bounded derivatives; (iv)* $K:\mathbb{R}\to[0,\infty)$ *is symmetric and satisfies* $K(u)=0$ *for all* $u\notin(-1,1)$*,* $\int K(u)du=1$*,* $\kappa^2:=\int u^2K(u)du\neq 0$*, and* $R_K:=\int K(u)^2du\in(0,\infty)$*.*

Note that Assumption NP allows for the most popular choices of symmetric and truncated kernels. To simplify notation, we define $k_t := K((x_t-x)/h)$ and $k_{tj} := K((x_t-x_j)/h)$. We also define the variance matrix

$$V := \begin{pmatrix} v^2 & \omega_{12} - v^2 \\ \omega_{12} - v^2 & v^2 + \omega_{22} - 2\omega_{12} \end{pmatrix},$$

where $v^2 := \sigma^2 R_K$, $\omega_{12} := \sigma^2 \int K(u) \int K(s-u)K(s)dsdu$, and $\omega_{22} := \sigma^2 \int (\int K(s-u)K(s)ds)^2 du$.

B.3.2 Proofs of (2.4) and lemmas

Although it is well known (e.g., Li and Racine, 2007) that (2.4) and Assumption 1 hold in this example, we give short proofs for completeness.

Proof of (2.4). Under Assumption NP we obtain by Taylor expansion the following well-known result,

$$\begin{aligned} E\hat{\beta}_h(x) &= \frac{1}{nh}\sum_{t=1}^{n} k_t \beta(x_t) = \int K(u)\beta(x+uh)du + o((nh)^{-1}) \\ &= \int K(u)\left(\beta(x) + \beta'(x)uh + \beta''(x)u^2h^2/2 + o(h^2)\right)du + o((nh)^{-1}) \\ &= \beta(x) + h^2\beta''(x)\kappa_2/2 + o(h^2) + o((nh)^{-1}), \end{aligned} \tag{B.10}$$

where the last equality follows by $\int K(u)du = 1$ and $\int uK(u)du = 0$. Setting the bandwidth as $h = cn^{-1/5}$ thus implies (2.4). Note that the limits of integration are $u \in ((n^{-1} - x)/h, (1-x)/h)$, but for n sufficiently large this is the same as $u \in (-1, 1)$ because $K(u) = 0$ for all $u \notin (-1, 1)$. We use this property throughout the remaining proofs.

Proof of Lemma 4.6. First, we verify Assumption 1 by showing that $\xi_{1,n} := T_n - B_n = (nh)^{-1/2}\sum_{t=1}^{n} k_t\varepsilon_t$ satisfies the central limit theorem. Because $k_t\varepsilon_t, t = 1, \ldots, n$, is a sequence of independent random variables with mean zero and $\mathrm{Var}(k_t\varepsilon_t) = k_t^2\sigma^2$, we have

$$\begin{aligned} \mathrm{Var}(\xi_{1,n}) &= \frac{1}{nh}\sum_{t=1}^{n} k_t^2\sigma^2 = \frac{\sigma^2}{h}\int K\left(\frac{s-x}{h}\right)^2 ds + o((nh)^{-1}) \\ &= \frac{\sigma^2}{h}\int K(u)^2 d(x+uh) + o((nh)^{-1}) \to \sigma^2 R_K = v^2. \end{aligned}$$

Moreover, Lyapunov's condition holds because

$$(nh)^{-(1+\delta)} \sum_{t=1}^{n} E(k_t^{2+\delta}|\varepsilon_t|^{2+\delta}) \leq c(nh)^{-(1+\delta)} \sum_{t=1}^{n} k_t^{2+\delta}$$
$$\leq c(nh)^{-(1+\delta)} \sum_{t:|x_t-x|\leq h} k_t^{2+\delta} \leq c(nh)^{-(1+\delta)} hn \to 0. \tag{B.11}$$

Next, we verify Assumption 2(i). Note that $T_n^* - \hat{B}_n = (nh)^{-1/2} \sum_{t=1}^{n} k_t \varepsilon_t^* =: \xi_{1,n}^*$, where, conditional on D_n, $\xi_{1,n}^* \sim N(0, \hat{\sigma}_n^2 (nh)^{-1} \sum_{t=1}^{n} k_t^2)$. Hence, the result follows from $\hat{\sigma}_n^2 \to_p \sigma^2$ and $(nh)^{-1} \sum_{t=1}^{n} k_t^2 \to R_K$.

Finally, we verify Assumption 2(ii). We first show that we can write

$$\hat{B}_n - B_n = \xi_{2,n} + o(1), \quad \xi_{2,n} := \frac{1}{\sqrt{nh}} \sum_{t=1}^{n} (\frac{1}{nh} \sum_{j=1}^{n} k_j k_{tj} - k_t) \varepsilon_t, \tag{B.12}$$

and then we show that

$$\xi_n := (\xi_{1,n}, \xi_{2,n})' \xrightarrow{d} N(0, V). \tag{B.13}$$

To prove (B.12) we write

$$\hat{B}_n - B_n = (nh)^{1/2} \left(\frac{1}{nh} \sum_{t=1}^{n} k_t (\hat{\beta}_h(x_t) - \beta(x_t)) - (\hat{\beta}_h(x) - \beta(x)) \right),$$

where

$$\hat{\beta}_h(x) = (nh)^{-1} \sum_{t=1}^{n} k_t \beta(x_t) + (nh)^{-1} \sum_{t=1}^{n} k_t \varepsilon_t,$$
$$k_t \hat{\beta}_h(x_t) = (nh)^{-1} \sum_{j=1}^{n} k_t k_{tj} \beta(x_j) + (nh)^{-1} \sum_{j=1}^{n} k_t k_{tj} \varepsilon_j.$$

By reversing the summations and exploiting symmetry of k_{tj}, it immediately follows that $\hat{B}_n - B_n = \xi_{2,n} + B_{2,n} - B_{1,n}$ with

$$B_{1,n}(x) := B_n = (nh)^{1/2} \left((nh)^{-1} \sum_{t=1}^{n} k_t \beta(x_t) - \beta(x) \right),$$
$$B_{2,n}(x) := (nh)^{-1} \sum_{t=1}^{n} k_t B_{1,n}(x_t) = (nh)^{-1/2} \sum_{t=1}^{n} k_t \left((nh)^{-1} \sum_{j=1}^{n} k_{tj} \beta(x_j) - \beta(x_t) \right).$$

By (B.10) we find

$$B_{2,n}(x) - B_{1,n}(x) = (nh)^{-1/2} \sum_{t=1}^{n} k_t \left(h^2 \beta''(x_t)\kappa_2/2 + o(h^2) + o((nh)^{-1}) \right)$$
$$- (nh)^{1/2} h^2 \beta''(x)\kappa_2/2 + o(h^2) + o((nh)^{-1})$$
$$= (nh)^{1/2} \frac{\kappa_2}{2} h^2 \left(\frac{1}{nh} \sum_{t=1}^{n} k_t \beta''(x_t) - \beta''(x) \right) + o((nh)^{1/2} h^2) + o((nh)^{-1/2}),$$

where

$$\frac{1}{nh} \sum_{t=1}^{n} k_t \beta''(x_t) = \int \beta''(x + uh) K(u) du + o((nh)^{-1}) = \beta''(x) + O(h) + o((nh)^{-1})$$

by first-order Taylor expansion, similar to (B.10), together with the assumption of continuous and bounded β'''. The result now follows because $h = cn^{-1/5}$.

Finally, to prove (B.13) we show that

$$J_n = \frac{1}{\sqrt{nh}} \sum_{t=1}^{n} \begin{pmatrix} k_t \\ (nh)^{-1} \sum_{j=1}^{n} k_j k_{tj} \end{pmatrix} \varepsilon_t \xrightarrow{d} N(0, \Omega), \quad \Omega := (\omega_{ij})_{i,j=1,2}, \tag{B.14}$$

from which the result follows by noting that $v^2 = \omega_{11}$ and

$$\xi_n = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} J_n.$$

It is clear that J_n has mean zero and independent increments. Approximating summations by integrals, it can be straightforwardly shown that

$$\mathrm{Var}(J_n) = \sigma^2 \begin{bmatrix} (nh)^{-1} \sum_{t=1}^{n} k_t^2 & (nh)^{-2} \sum_{t,j=1}^{n} k_t k_j k_{tj} \\ (nh)^{-2} \sum_{t,j=1}^{n} k_t k_j k_{tj} & (nh)^{-1} \sum_{t=1}^{n} ((nh)^{-1} \sum_{j=1}^{n} k_j k_{tj})^2 \end{bmatrix} \to \Omega.$$

By the same proof as in (B.11), we can show that the Lyapunov condition is satisfied, and result (B.14) follows. □

Proof of Lemma 4.7. We first verify Assumption 3(i). We notice that $\bar{T}_n^{**} - \bar{B}_n^* = T_n^{**} - \hat{B}_n^* = (nh)^{-1/2} \sum_{t=1}^{n} k_t \varepsilon_i^{**} =: \xi_{1,n}^{**}$, where, conditional on (D_n, D_n^*), $\xi_{1,n}^{**} \sim N(0, \hat{\sigma}_n^{*2} (nh)^{-1} \sum_{t=1}^{n} k_t^2)$. Hence, the result follows from $\hat{\sigma}_n^{*2} \xrightarrow{p^*}_p \sigma^2$ and $(nh)^{-1} \sum_{t=1}^{n} k_t^2 \to R_K$.

Next, we verify Assumption 3(ii). We first write $T_n^* - \bar{B}_n^* = T_n^* - \hat{B}_n - (\bar{B}_n^* - \hat{B}_n) = \xi_{1,n}^* - (\bar{B}_n^* - \hat{B}_n)$, where $\bar{B}_n^* - \hat{B}_n = \hat{B}_n^* - \hat{B}_{2,n}$. Recall $\hat{B}_{2,n} := (nh)^{-1}\sum_{t=1}^n k_t \hat{B}_n(x_t) = (nh)^{-1/2}\sum_{t=1}^n k_t((nh)^{-1}\sum_{j=1}^n k_{tj}\hat{\beta}_h(x_j) - \hat{\beta}_h(x_t))$ and $\hat{B}_n^* := (nh)^{1/2}((nh)^{-1}\sum_{t=1}^n k_t\hat{\beta}_h^*(x_t) - \hat{\beta}_h^*(x))$, where

$$\hat{\beta}_h^*(x) = (nh)^{-1}\sum_{t=1}^n k_t\hat{\beta}_h(x_t) + (nh)^{-1}\sum_{t=1}^n k_t\varepsilon_t^*,$$
$$k_t\hat{\beta}_h^*(x_t) = (nh)^{-1}\sum_{j=1}^n k_t k_{tj}\hat{\beta}_h(x_j) + (nh)^{-1}\sum_{j=1}^n k_t k_{tj}\varepsilon_j^*,$$

so it follows that

$$\bar{B}_n^* - \hat{B}_n = \hat{B}_n^* - \hat{B}_{2,n} = \xi_{2,n}^*, \quad \xi_{2,n}^* := \frac{1}{\sqrt{nh}}\sum_{t=1}^n\Big(\frac{1}{nh}\sum_{j=1}^n k_j k_{tj} - k_t\Big)\varepsilon_t^*.$$

Thus, the proof is completed by showing that

$$\xi_n^* := (\xi_{1,n}^*, \xi_{2,n}^*)' \xrightarrow{d^*}_p N(0, V). \tag{B.15}$$

Conditional on D_n, it holds that $\xi_n^* \sim N(0, \hat{V}_n)$, where

$$\hat{V}_n = \hat{\sigma}_n^2 \frac{1}{nh}\sum_{t=1}^n \begin{bmatrix} k_t^2 & k_t(\frac{1}{nh}\sum_{j=1}^n k_j k_{tj} - k_t) \\ k_t(\frac{1}{nh}\sum_{j=1}^n k_j k_{tj} - k_t) & (\frac{1}{nh}\sum_{j=1}^n k_j k_{tj} - k_t)^2 \end{bmatrix} \xrightarrow{p} V$$

by approximating the summations by integrals and using $\hat{\sigma}_n^2 \to_p \sigma^2$. This proves (B.15) and hence completes the proof of Lemma 4.7. $\square$

B.4 Inference under heavy tails

Setup. We consider a simple location model with heavy-tailed data, thus demonstrating that our analysis applies to a non-Gaussian asymptotic framework. Specifically, consider a sample of n i.i.d. random variables $\{y_t\}$. Interest is in inference on θ in the location model

$$y_t = \theta + \varepsilon_t, \quad E(\varepsilon_t) = 0,$$

when the ε_t's follow a symmetric, stable random variable $S(\alpha)$ with tail index $\alpha \in (1,2)$ and the location parameter is local to zero; i.e., $\theta = n^{1/\alpha-1}c$.[4] Under these assumptions, $E(|\varepsilon_t|^{\alpha+\delta}) = +\infty$ for any $\delta \geq 0$; in particular, ε_t has infinite variance. Notice that θ is local of order $n^{1/\alpha-1}$ rather than the usual $n^{-1/2}$ because of the slower convergence rate of the OLS-type estimator when the variance of ε_t is infinite. We consider the biased estimator

$$\hat{\theta}_n := \omega \bar{y}_n, \quad \bar{y}_n := n^{-1} \sum_{t=1}^{n} y_t,$$

where $\omega \in (0,1)$. In the finite variance case, this estimator improves upon $\bar{y}_n$ in terms of MSE when θ is local to zero. It holds that

$$T_n := n^{1-1/\alpha}(\hat{\theta}_n - \theta) = (\omega - 1)c + \omega n^{1-1/\alpha} \bar{\varepsilon}_n \sim B + \omega S(\alpha) \tag{B.16}$$

with $B := (\omega - 1)c$; equivalently, $T_n - B \sim \xi_1 := \omega S(\alpha)$. Hence, Assumption 1 is satisfied with $G(u) = P(\omega S(\alpha) \leq u) = \Psi_\alpha(\omega^{-1}u)$, where $\Psi_\alpha(u) := P(S(\alpha) \leq u)$ is continuous. Inference based on quantiles of ξ_1 is invalid because it misses the term B.

Bootstrap. It is well known that the standard bootstrap fails to be valid under infinite variance (Knight, 1989). The 'm out of n' bootstrap (see Politis et al., 1999, and the references therein) is an attractive option, but it fails to mimic the non-centrality parameter B; see Remark B.2 below. Instead, we consider the parametric bootstrap of Cornea-Madeira and Davidson (2015), which only requires a consistent estimator $\hat{\alpha}_n$ of the tail index α, assumed to lie in a compact set. The bootstrap sample is generated as

$$y_t^* = \bar{y}_n + \varepsilon_t^*, \quad \varepsilon_t^* \sim \text{ i.i.d.} S(\hat{\alpha}_n),$$

and the bootstrap estimator is $\hat{\theta}_n^* := \omega \bar{y}_n^* = \omega(\bar{y}_n + \bar{\varepsilon}_n^*)$ with $\bar{\varepsilon}_n^* := n^{-1} \sum_{t=1}^{n} \varepsilon_t^*$. The bootstrap analogue of T_n then satisfies

$$T_n^* := n^{1-1/\alpha}(\hat{\theta}_n^* - \bar{y}_n) = \omega n^{1-1/\alpha} \bar{\varepsilon}_n^* + \hat{B}_n \text{ with } \hat{B}_n := (\omega - 1) n^{1-1/\alpha} \bar{y}_n.$$

[4] The results in this section can easily be generalized to the case where the ε_t's are not necessarily symmetric and/or are in the domain of attraction of a stable law with index $\alpha \in (0,1)$, as in Cornea-Madeira and Davidson (2015). Moreover, the results apply to the case of non-local θ as well; i.e., $\theta \neq 0$ fixed.

Now, $n^{1-1/\alpha}\bar{\varepsilon}_n^* \xrightarrow{d^*}_p S(\alpha)$ by Proposition 1 in Cornea-Madeira and Davidson (2015) and, therefore,

$$T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1 := \omega S(\alpha).$$

This shows that Assumption 2(i) is satisfied in this example. Notice that the bias term in the bootstrap world satisfies, jointly with (B.16),

$$\hat{B}_n - B = (\omega - 1)n^{1-1/\alpha}\bar{\varepsilon}_n \sim (\omega - 1)S(\alpha) =: \xi_2.$$

Specifically, because both T_n and $\hat{B}_n$ depend on the data through $\bar{\varepsilon}_n$ only, we have that $(\xi_1, \xi_2) \sim (\omega, \omega - 1)S(\alpha)$, implying that $\xi_1 - \xi_2 \sim S(\alpha)$. Hence, Assumption 2(ii) is satisfied with $F(u) = P(S(\alpha) \leq u) = \Psi_\alpha(u)$. Since the cdf of $\xi_1 \sim \omega S(\alpha)$ can be written as $G(u) = \Psi_\alpha(\omega^{-1}u)$, it follows by Theorem 3.1 that $\hat{p}_n \to_d G(F^{-1}(U_{[0,1]})) = \Psi_\alpha(\omega^{-1}\Psi_\alpha^{-1}(U_{[0,1]}))$ and, therefore,

$$P(\hat{p}_n \leq u) \to H(u) := P(\Psi_\alpha(\omega^{-1}\Psi_\alpha^{-1}(U_{[0,1]})) \leq u) = \Psi_\alpha(\omega\Psi_\alpha^{-1}(u)),$$

which differs from u unless $\omega = 1$.

Because ω is known and we can estimate α consistently with $\hat{\alpha}_n$, we can estimate $H(u)$ consistently with $\hat{H}_n(u) := \Psi_{\hat{\alpha}_n}(\omega\Psi_{\hat{\alpha}_n}^{-1}(u))$ and obtain a valid plug-in modified p-value,

$$\tilde{p}_n = \hat{H}_n(\hat{p}_n) = \Psi_{\hat{\alpha}_n}(\omega\Psi_{\hat{\alpha}_n}^{-1}(\hat{p}_n)),$$

by application of Corollary 3.2.

Alternatively, we can estimate $H(u)$ using the double bootstrap estimator $\hat{H}_n(u) := P^*(\hat{p}_n^{**} \leq u)$, where $\hat{p}_n^{**} := P^{**}(T_n^{**} \leq T_n^*)$. Specifically, let the double bootstrap sample $\{y_t^{**}\}$ be generated as

$$y_t^{**} = \bar{y}_n^* + \varepsilon_t^{**}, \quad \varepsilon_t^{**} \sim \text{ i.i.d.} S(\hat{\alpha}_n),$$

and set $\hat{\theta}_n^{**} := \omega\bar{y}_n^{**} = \omega\bar{y}_n^* + \omega\bar{\varepsilon}_n^{**}$, where $\bar{\varepsilon}_n^{**} := n^{-1}\sum_{t=1}^n \varepsilon_t^{**}$. The (second-level) bootstrap analogue of T_n^* then satisfies

$$T_n^{**} := n^{1-1/\alpha}(\hat{\theta}_n^{**} - \bar{y}_n^*) = \omega n^{1-1/\alpha}\bar{\varepsilon}_n^{**} + \hat{B}_n^* \text{ with } \hat{B}_n^* := (\omega - 1)n^{1-1/\alpha}\bar{y}_n^*.$$

Since ε_t^{**} is generated from $S(\hat{\alpha}_n)$, where $\hat{\alpha}_n$ depends only on D_n, the distribution of ε_t^{**}, conditionally on D_n^* and D_n, is the same as the distribution of ε_t^*, conditionally on D_n. This implies that

$$n^{1-1/\alpha}\bar{\varepsilon}_n^{**} \xrightarrow{d^{**}}_{p*} S(\alpha),$$

in probability, by Proposition 1 of Cornea-Madeira and Davidson (2015). Therefore,

$$T_n^{**} - \hat{B}_n^* \xrightarrow{d^{**}}_{p*} \xi_1 = \omega S(\alpha),$$

in probability, showing that Assumption 3(i) is satisfied. Since

$$\hat{B}_n^* - \hat{B}_n = (\omega - 1)n^{1-1/\alpha}(\bar{y}_n^* - \bar{y}_n) = (\omega - 1)n^{1-1/\alpha}\bar{\varepsilon}_n^*$$

and $T_n^* - \hat{B}_n = \omega n^{1-1/\alpha}\bar{\varepsilon}_n^*$, Assumption 3(ii) is also satisfied in this example. Thus, $\tilde{p}_n = \hat{H}_n(\hat{p}_n) \to_d U_{[0,1]}$ by Theorem 3.2.

Remark B.2 *Consider the 'm out of n' bootstrap data generating process,*

$$y_t^* = \bar{y}_n + \varepsilon_t^*, \quad t = 1, \ldots, m,$$

where ε_t^* is an i.i.d. sample from the residuals $\hat{\varepsilon}_t = y_t - \bar{y}_n$, $t = 1, \ldots, n$. Then, with $\hat{\theta}_m^* := \omega \bar{y}_m^*$, $\bar{y}_m^* := m^{-1}\sum_{t=1}^m y_t^*$, the 'm out of n' bootstrap statistic is

$$T_m^* := m^{1-1/\alpha}(\hat{\theta}_m^* - \bar{y}_n) = \omega m^{1-1/\alpha}\bar{\varepsilon}_m^* + (\omega - 1)m^{1-1/\alpha}\bar{y}_n,$$

where $m^{1-1/\alpha}\bar{\varepsilon}_m^* \xrightarrow{d^*}_p S(\alpha)$ as $m \to \infty$; see Arcones and Giné (1989). Moreover, if $m = o(n)$,

$$\hat{B}_m := (\omega - 1)m^{1-1/\alpha}\bar{y}_n = (\omega - 1)m^{1-1/\alpha}n^{1/\alpha-1}(n^{1-1/\alpha}\bar{y}_n) = O_p((m/n)^{1/\alpha-1}) = o_p(1),$$

which shows that $T_m^* \xrightarrow{d^*}_p \omega S(\alpha)$. Hence, Assumption 2(i) is satisfied with $\xi_1 := \omega S(\alpha)$ and $\hat{B}_n = 0$. Since $B := (\omega - 1)c \neq 0$, we have $\xi_2 := -B$ a.s., so that Assumption 2(ii) does not hold. As in Remark 3.1, it then follows that

$$\hat{p}_m := P^*(T_m^* \leq T_n) \xrightarrow{d} G(G^{-1}(U_{[0,1]}) + B) = \Psi_\alpha(\Psi_\alpha^{-1}(U_{[0,1]}) + B).$$

This shows that the limiting distribution of $\hat{p}_m$ depends on B. Since B cannot be consistently estimated, *the 'm out of n' bootstrap cannot be used to solve the problem.*

B.5 Nonlinear dynamic panel data models with incidental parameter bias

Another example that fits our framework is inference based on panel data estimators subject to incidental parameter bias. We consider the properties of the cross-sectional pairs bootstrap considered by Kaffo (2014), Dhaene and Jochmans (2015), and Gonçalves and Kaffo (2015) in the context of a general nonlinear panel data model. Although this bootstrap cannot replicate the bias, we show that our prepivoting approach based on a plug-in estimator of the bias is valid. Recently, Higgins and Jochmans (2022) proposed a (double) bootstrap procedure that retains asymptotic validity without an explicit plug-in estimator of the bias, but their procedure relies heavily on the parametric distribution assumption.

Setup. Let z_{it} denote a vector of random variables for a set of n individuals, $i = 1, \ldots, n$, over T time periods, $t = 1, \ldots, T$. Given a model for the density function $f_{it}(\theta, \alpha_i) := f(z_{it}, \theta, \alpha_i)$, the parameter of interest is $\theta \in \Theta$, which is common to all the individuals, while $\alpha_i \in \mathcal{A}$ denote the individual fixed effects. The fixed effects estimator of θ is the maximum likelihood estimator defined as

$$\hat{\theta}_n = \arg\max_{\theta\in\Theta} \sum_{i=1}^{n}\sum_{t=1}^{T} \log f_{it}(\theta, \hat{\alpha}_i(\theta)), \text{ where } \hat{\alpha}_i(\theta) = \arg\max_{\alpha_i\in\mathcal{A}} \sum_{t=1}^{T} \log f_{it}(\theta, \alpha_i). \tag{B.17}$$

Under certain regularity conditions (see, e.g., Hahn and Kuersteiner, 2011), including letting $n, T \to \infty$ jointly such that $n/T \to \rho < \infty$,

$$T_n := \sqrt{nT}(\hat{\theta}_n - \theta) \xrightarrow{d} N(B, v^2), \tag{B.18}$$

where B denotes the incidental parameter bias and v^2 is the asymptotic variance of $\hat{\theta}_n$. Hence, Assumption 1 is satisfied with $\xi_1 \sim N(0, v^2)$ (equivalently, Assumption $1'$ is satisfied).

The exact forms of B and v^2 may be quite involved and depend on the type of heterogeneity and dependence assumptions imposed on z_{it}. A standard assumption is that z_{it} is independent across i while allowing for time series dependence of unknown form; see Hahn and Kuersteiner (2011).

Bootstrap. Given the cross-sectional independence assumption, a natural bootstrap method in this context is the cross-sectional pairs bootstrap. The idea is to resample $z_i = (z_{i1}, \ldots, z_{iT})'$ in an i.i.d. fashion in the cross-sectional dimension. If $z_{it} = (y_{it}, x_{it})'$ and $f(z_{it}, \theta, \alpha_i) = f(y_{it}|x_{it}, \theta, \alpha_i)$ is the conditional density of y_{it} given x_{it}, this is equivalent to a cross-sectional pairs bootstrap. As the results of Kaffo (2014, Theorem 3.1) show, this bootstrap fails to capture the bias term B. In particular, letting $\hat{\theta}_n^*$ denote the bootstrap analogue of $\hat{\theta}_n$, we have that

$$T_n^* := \sqrt{nT}(\hat{\theta}_n^* - \hat{\theta}_n) \xrightarrow{d^*}_p N(0, v^2),$$

which implies that, as in Remarks 3.1 and B.2,

$$\hat{p}_n := P^*(T_n^* \leq T_n) = \Phi(v^{-1}T_n) + o_p(1) \xrightarrow{d} \Phi(v^{-1}B + \Phi^{-1}(U_{[0,1]})).$$

Thus,

$$P(\hat{p}_n \leq u) \to H(u) := P(\Phi(\Phi^{-1}(U_{[0,1]}) + v^{-1}B) \leq u) = \Phi(\Phi^{-1}(u) - v^{-1}B),$$

which shows that the bootstrap test based on $\hat{p}_n$ is asymptotically invalid since its limiting distribution is not uniform.

Remark B.3 *Note that, in this example, $\hat{L}_n(u) := P^*(T_n^* \leq u) \to_p \Phi(u/v)$, showing that the bootstrap conditional distribution of T_n^* is not random in the limit. The invalidity of $\hat{p}_n$ is due to the fact that the cross-sectional pairs bootstrap induces $\hat{B}_n = 0$, whereas $B \neq 0$. This implies that $\hat{B}_n - B = -B := \xi_2$ is not random. The fact that ξ_2 is not zero is the cause of the bootstrap invalidity. See Remark 3.1, which contains this example as a special case.*

Contrary to previous examples (e.g., Remark B.2), B and v can both be consistently estimated. Hence, in this example we can restore bootstrap validity by modifying the bootstrap p-value using a plug-in approach. More specifically, let $\tilde{B}_n$ and $\hat{v}_n$ denote consistent estimators of B and v, respectively.[5]

[5] Since we reserve the notation $\hat{B}_n$ for the bootstrap-induced bias estimator (which is zero for the cross sectional pairs bootstrap), we use the notation $\tilde{B}_n$ to denote any consistent estimator of B in this setup. For instance, $\tilde{B}_n$ could be the plug-in estimator proposed by Hahn and Kuersteiner (2011), which is based on a closed-form expression of B_1. Another option is the half-split panel jackknife estimator of Dhaene and Jochmans (2015).

By Corollary 3.2,

$$\tilde{p}_n = \hat{H}_n(\hat{p}_n) = \Phi(\Phi^{-1}(\hat{p}_n) - \hat{v}_n^{-1}\tilde{B}_n) \xrightarrow{d} U_{[0,1]}$$

because $\hat{H}_n(u) := \Phi(\Phi^{-1}(u) - \hat{v}_n^{-1}\tilde{B}_n)$ is a consistent estimator of $H(u)$.

Remark B.4 *A double bootstrap modified p-value version of $\tilde{p}_n$ is not valid in this setting. The reason is that the double bootstrap mimics the behavior of the first-level bootstrap, i.e.*

$$T_n^{**} := \sqrt{nT}(\hat{\theta}_n^{**} - \hat{\theta}_n^{*}) \xrightarrow{d^{**}}_p N(0, v^2),$$

so that $\hat{B}_n^*$ in Assumption 3(i) is zero. Since $\hat{B}_n = 0$, Assumption 3(ii) holds with $\hat{B}_n^* - \hat{B}_n = 0$, whereas Assumption 2(ii) has $\hat{B}_n - B_n = -B$ a.s. Then,

$$\hat{p}_n^* = P^{**}(v^{-1}T_n^{**} \leq v^{-1}T_n^*) = \Phi(v^{-1}T_n^*) \xrightarrow{d^*}_p \Phi(\Phi^{-1}(U_{[0,1]})) = U_{[0,1]},$$

whereas

$$\hat{p}_n \xrightarrow{d} \Phi(\Phi^{-1}(U_{[0,1]}) + v^{-1}B).$$

Thus, $\hat{H}_n(u) := P^*(\hat{p}_n^* \leq u)$ is not a consistent estimator of $H(u)$, invalidating $\tilde{p}_n = \hat{H}_n(\hat{p}_n)$.

Remark B.5 *A special case of the previous setup is a linear panel dynamic model, where $z_{it} = (y_{it}, x_{it}')'$ and x_{it} is a vector containing lags of y_{it} (Hahn and Kuersteiner, 2002). In this case, the plug-in modified p-value, $\tilde{p}_n$, based on the cross-sectional pairs bootstrap can be implemented using any consistent estimator of B, as described above. However, we can also use a recursive bootstrap that exploits the linearity of the model to obtain an asymptotically valid standard bootstrap p-value, $\hat{p}_n$. The validity of $\hat{p}_n$ follows from the fact that the recursive bootstrap estimates B consistently, contrary to the pairs bootstrap (Gonçalves and Kaffo, 2015). In light of this, prepivoting $\hat{p}_n$ by computing a double bootstrap modified p-value $\tilde{p}_n = \hat{H}_n(\hat{p}_n)$ is not needed in this example, but it is still a valid alternative.*

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